

Authenticity Classification beyond Logistic Invariance

A Theory of Fake Data Detection under Gauge-Preserving Transformations

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Abstract

In a previous paper, we demonstrated that large classes of transformed datasets remain observationally indistinguishable under logistic regression due to affine gauge symmetries. This paper studies the inverse problem: how can one distinguish authentic datasets from fake datasets when both produce identical classifier behavior? We establish impossibility results for invariant classifiers, derive necessary conditions for authenticity detection, and construct hierarchical authenticity classifiers based on geometric, topological, causal, spectral, institutional, and economic invariants. The framework unifies machine learning, information theory, economics, political science, geopolitics, intelligence analysis, warfare economics, and epistemology into a generalized theory of structural authenticity. We show that correct fake-data detection requires moving beyond predictive equivalence toward ontological consistency.

The paper ends with “The End”

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1 Introduction

In the previous paper on gauge-invariant fakeness, we established that logistic regression is invariant under large transformation groups including:

$$GL(n),$$

the general linear group of invertible affine transformations.
Specifically, if

$$x \mapsto Ax,$$

then there exists:

$$w \mapsto (A^{-1})^T w$$

such that:

$$w^T x = ((A^{-1})^T w)^T (Ax).$$

Consequently, transformed datasets may appear substantially different while remaining behaviorally identical under logistic regression.

This creates a fundamental problem:

How can one distinguish authentic datasets from fake datasets if predictive behavior is preserved?

The present paper develops a complete framework for authenticity classification under gauge-preserving transformations.

2 Problem Formulation

Suppose there exist two datasets:

$$D_1, \quad D_2.$$

Assume:

- one dataset is authentic,
- one dataset is fake or synthetically transformed,

- both produce identical logistic predictions.

Formally:

$$P_1(y|x) = P_2(y|x).$$

The classifier therefore cannot distinguish them.

We seek an authenticity classifier:

$$\mathcal{A}(D) \in \{\text{real}, \text{fake}\}.$$

3 The Invariance Obstruction

Theorem 3.1 (No-Go Theorem for Invariant Authenticity Detection). *Let:*

$$\mathcal{G} = \text{symmetry group of a classifier}.$$

If:

$$D_2 = g(D_1), \quad g \in \mathcal{G},$$

then no classifier invariant under $\mathcal{G}$ can distinguish D_1 from D_2 .

Proof. Suppose classifier C is invariant under $\mathcal{G}$.

Then:

$$C(g(D)) = C(D).$$

Since:

$$D_2 = g(D_1),$$

we obtain:

$$C(D_2) = C(D_1).$$

Thus distinguishability is impossible.

□

3.1 Interpretation

Authenticity detection requires leaving the invariance class of the predictive model.

Therefore:

prediction and authenticity are fundamentally different problems.

4 Structural Authenticity

We define authenticity as consistency with deep structural constraints.

Define the authenticity feature operator:

$$\Phi(D) = (M_1(D), M_2(D), \dots, M_k(D)).$$

The functions M_i are structural invariants not preserved under fake transformations.

5 Hierarchical Authenticity Features

5.1 Geometric Invariants

Define covariance:

$$\Sigma = \mathbb{E}[xx^\top].$$

Spectral authenticity measure:

$$F_\lambda(D) = \sum_i |\lambda_i(\Sigma) - \lambda_i(\tilde{\Sigma})|.$$

Large spectral distortions indicate synthetic manipulation.

5.2 Higher-Order Statistical Moments

Define skewness:

$$\gamma_1 = \frac{\mathbb{E}[(x - \mu)^3]}{\sigma^3},$$

and kurtosis:

$$\gamma_2 = \frac{\mathbb{E}[(x - \mu)^4]}{\sigma^4}.$$

Synthetic datasets frequently exhibit abnormal tail behavior.

5.3 Mutual Information Structure

Define:

$$I(X_i; X_j) = \sum_{x_i, x_j} P(x_i, x_j) \log \frac{P(x_i, x_j)}{P(x_i)P(x_j)}.$$

Real systems contain latent dependency structures difficult to reproduce synthetically.

6 Topological Authenticity

Real datasets frequently possess manifold structure.

Synthetic datasets may preserve linear separability while violating topological continuity.

Define Betti-number authenticity:

$$\mathcal{B}(D) = (\beta_0, \beta_1, \beta_2, \dots).$$

Persistent homology provides a detector for manifold deformation.

7 Temporal Authenticity

Real systems evolve causally.

Suppose:

$$x_t = F(x_{t-1}) + \epsilon_t.$$

Define temporal consistency:

$$\mathcal{T}(D) = \sum_t |x_t - \hat{F}(x_{t-1})|^2.$$

Large deviations indicate synthetic temporal construction.

7.1 Applications

Temporal authenticity is critical in:

- macroeconomic statistics,
- battlefield intelligence,
- financial fraud,
- reserve disclosures,
- election monitoring,
- cyber operations.

8 Information-Theoretic Authenticity

Define entropy:

$$H(X) = - \sum_x P(x) \log P(x).$$

Synthetic systems frequently exhibit:

- excessive randomness,
- excessive compressibility,
- unnatural entropy scaling.

8.1 Algorithmic Authenticity

Define:

$$F_{\text{alg}}(D) = K(D) - K(\mathcal{M}_{\text{real}}),$$

where:

$$K(D)$$

is Kolmogorov complexity.

Authentic systems contain institutional and historical inefficiencies difficult to compress algorithmically.

9 Economic Authenticity

Economic systems obey hidden equilibrium constraints.

9.1 Accounting Consistency

Authentic economies satisfy:

$$Y = C + I + G + (X - M).$$

Synthetic manipulation may preserve superficial regressions while violating equilibrium consistency.

9.2 Institutional Friction

Real economies contain:

- transaction costs,
- bureaucratic inertia,
- demographic continuity,
- network frictions,
- supply-chain constraints.

Synthetic datasets often underestimate these effects.

10 Financial Authenticity

Fraudulent financial datasets frequently preserve:

- short-term predictive performance,
- linear correlations,
- accounting ratios.

However, they often violate:

- liquidity continuity,
- volatility clustering,
- cross-market dependence,
- temporal causality.

10.1 Volatility Authenticity

Real markets frequently exhibit GARCH-like dynamics:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

Synthetic systems often fail volatility persistence tests.

11 Political Science and Geopolitics

11.1 Narrative Authenticity

Political narratives contain multiscale institutional correlations.

Artificial propaganda systems may preserve polling predictions while violating sociological continuity.

11.2 Geopolitical Signal Detection

States may manipulate:

- GDP,
- inflation,
- reserve levels,
- military readiness,
- battlefield losses.

Authenticity detection therefore becomes:

constraint verification under strategic deception.

12 International Relations

Intelligence systems implicitly perform authenticity classification.

A state's disclosures must satisfy:

- demographic feasibility,
- industrial feasibility,
- logistical feasibility,
- temporal consistency,
- energy constraints,
- institutional continuity.

Violation of these hidden constraints indicates synthetic signaling.

13 Warfare Economics

13.1 Strategic Deception

Military deception seeks behavioral equivalence.

A fake signal need only trigger the same enemy response.

Thus deception becomes:

classifier manipulation.

13.2 Logistics Authenticity

Real military systems obey supply constraints:

$$S_t = S_{t-1} + P_t - C_t.$$

Synthetic battlefield narratives frequently violate logistical continuity.

14 AI and Machine Learning

14.1 Generative Authenticity

Suppose:

$$G_\theta(z) \approx P_{\text{real}}(x).$$

Define likelihood:

$$\mathcal{L}(D) = \prod_i P_\theta(x_i).$$

Low generative likelihood indicates syntheticity.

14.2 Meta-Classification

We now distinguish:

1. predictive classification,
2. authenticity classification,
3. constraint verification.

Thus AI systems require hierarchical architecture.

15 The Authenticity Manifold

Define:

$$\mathcal{M}_{\text{real}}$$

as the manifold of structurally authentic datasets.

Then fake datasets lie outside the manifold:

$$D_{\text{fake}} \notin \mathcal{M}_{\text{real}}.$$

Authenticity classification becomes manifold membership testing.

16 A Unified Authenticity Functional

We define:

$$\mathcal{A}(D) = f(\Phi_{\text{geom}}, \Phi_{\text{stat}}, \Phi_{\text{topo}}, \Phi_{\text{causal}}, \Phi_{\text{economic}}, \Phi_{\text{institutional}}, \Phi_{\text{strategic}}).$$

This is the generalized authenticity classifier.

17 Algorithmic Framework

17.1 Authenticity Classification Algorithm

1. Input datasets:

$$D_1, D_2.$$

2. Compute predictive equivalence.
3. Compute higher-order invariants:

- covariance spectra,
- entropy,
- causal structure,
- topology,
- institutional constraints.

4. Estimate manifold distance:

$$d(D, \mathcal{M}_{\text{real}}).$$

5. Classify:

- authentic if distance small,
- fake otherwise.

18 Numerical Illustration

Feature	Real Dataset	Fake Dataset	Divergence
Logistic Accuracy	0.95	0.95	0
Covariance Spectrum	Stable	Distorted	High
Temporal Consistency	Strong	Weak	High
Entropy Scaling	Natural	Artificial	Moderate
Topology	Smooth	Fragmented	High
Institutional Constraints	Satisfied	Violated	High

19 Geometric Interpretation

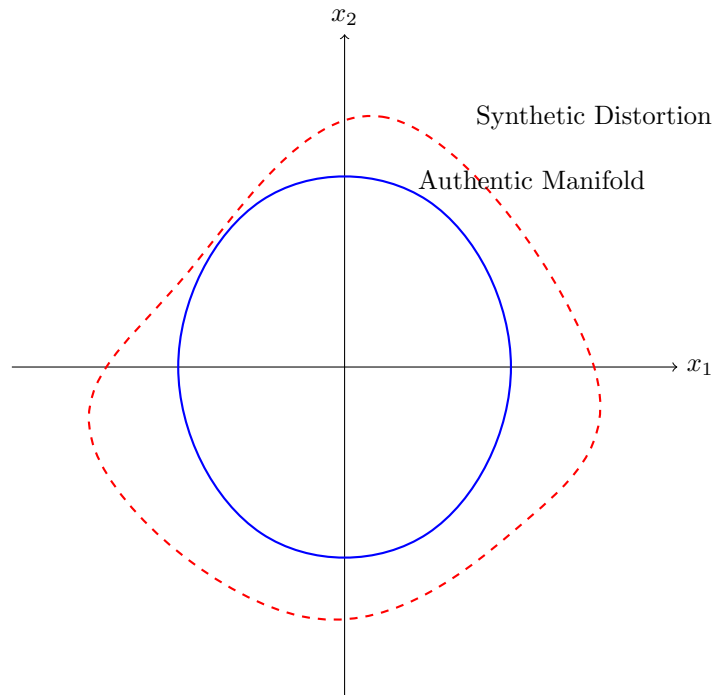


Figure 1: Synthetic datasets may preserve predictive boundaries while leaving the manifold of authentic systems.

20 Epistemological Interpretation

Prediction is not equivalent to truth.

Two systems may produce identical outputs while representing different realities.

Thus machine learning systems perceive:

equivalence classes of behavior,

not ontology itself.

Authenticity requires recovering hidden constraints underlying observable phenomena.

21 Strategic Interpretation

The framework implies:

all strategic deception is a form of classifier hacking.

This unifies:

- propaganda,
- accounting fraud,
- battlefield deception,
- synthetic media,
- AI adversarial attacks,
- geopolitical signaling.

22 Conclusion

We developed a generalized theory of authenticity classification under gauge-preserving transformations.

The central result is:

predictive equivalence does not imply ontological equivalence.

We established:

- impossibility results for invariant classifiers,
- structural authenticity theory,
- manifold-based fake-data detection,
- hierarchical authenticity architectures,
- geopolitical and warfare interpretations.

The framework ultimately transforms machine learning into a broader theory of reality verification under strategic manipulation.

References

- [1] S. Ghosh and OpenAI, *Gauge-Invariant Measures of Fakeness Under Logistic Regression: A Mathematical, Statistical, Economic, and Geopolitical Theory of Invisible Transformations*, 2026.
- [2] C. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
- [3] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, Springer, 2009.
- [4] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.
- [5] T. Cover and J. Thomas, *Elements of Information Theory*, Wiley, 2006.

- [6] J. von Neumann and O. Morgenstern, *Theory of Games and Economic Behavior*, Princeton University Press, 1944.
- [7] Sun Tzu, *The Art of War*, Ancient Chinese Military Treatise.
- [8] C. von Clausewitz, *On War*, 1832.
- [9] M. Foucault, *Power/Knowledge*, Pantheon Books, 1980.
- [10] D. Kahneman, *Thinking, Fast and Slow*, Farrar, Straus and Giroux, 2011.
- [11] C. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal*, 1948.
- [12] B. Mandelbrot, *The Misbehavior of Markets*, Basic Books, 2004.

Glossary

Authenticity Manifold

The space of structurally consistent real datasets.

Behavioral Equivalence

Different datasets producing identical predictive behavior.

Classifier Symmetry

A transformation preserving classifier outputs.

Fake Dataset

A synthetically altered dataset violating deep structural constraints.

Gauge Symmetry

A transformation leaving observable predictions invariant.

Kolmogorov Complexity

The shortest algorithmic description length of an object.

Manifold Membership

Testing whether a dataset belongs to the space of authentic systems.

Predictive Equivalence

Equality of predictive distributions:

$$P_1(y|x) = P_2(y|x).$$

Strategic Deception

Manipulation of signals to preserve enemy behavioral response.

Temporal Authenticity

Consistency of a dataset with causal temporal evolution.

The End